Advanced AFM Applications Training

PeakForce-QNM II: QNM Calibrations

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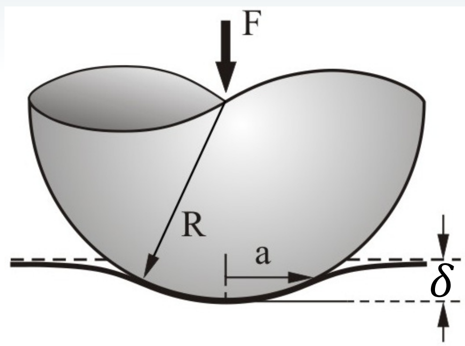
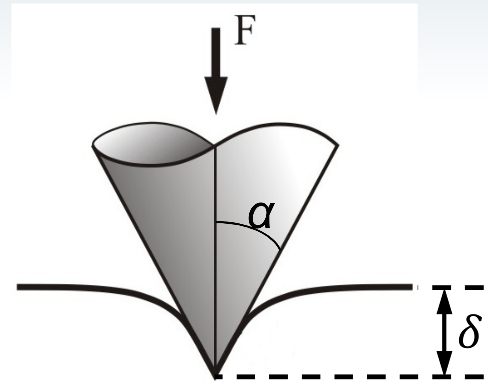
Calibration Parameters





$$F = \frac{4}{3}\frac{E}{(1-\nu^2)}R^{1/2}\delta^{3/2}$$

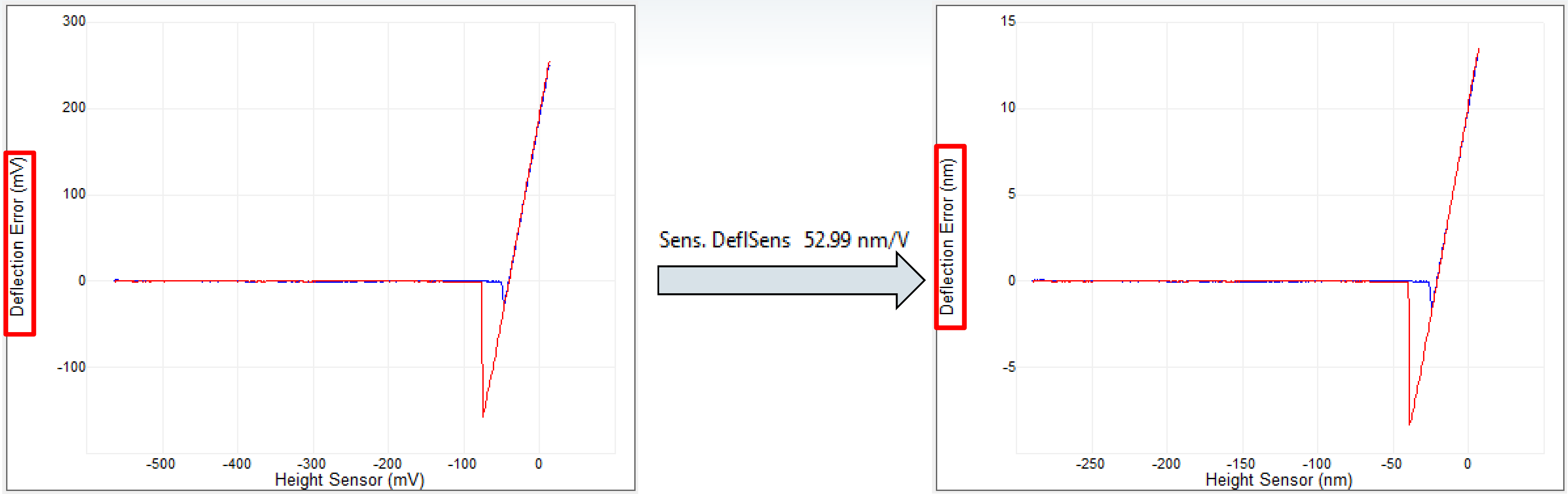
$$F = \frac{2}{\pi}\frac{E}{(1-\nu^2)}\tan(\alpha)\delta^2$$

- **Deflection Sensitivity**
- **Spring Constant**
- **Tip Radius or tip half angle**
- Sample Poisson's Ratio (material property, typically 0.2-0.5)

- **Question:** which of the above parameters is the most critical calibration parameter?

1. **Deflection Sensitivity**
2. Spring Constant
3. Tip Radius

What is Deflection Sensitivity

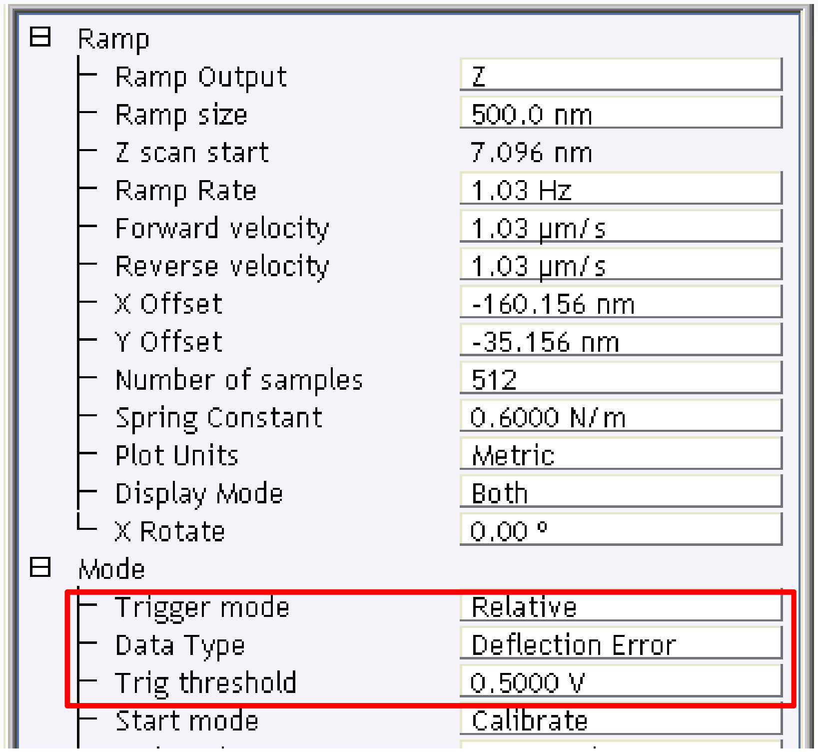


- AFM reads cantilever deflection in **V**
- Deflection Sensitivity converts deflection in **V** into deflection in **nm**
- Deflection Sensitivity unit: **nm/V**
- Deflection Sensitivity depends on: **1.** cantilever type, **2.** instrument type, **3.** laser alignment, **4.** surrounding media (air or fluid)

- **Question:** 50nm/V and 100nm/V, which one is better Deflection Sensitivity?
- **Question:** Shorter or longer cantilever has better Deflection Sensitivity?

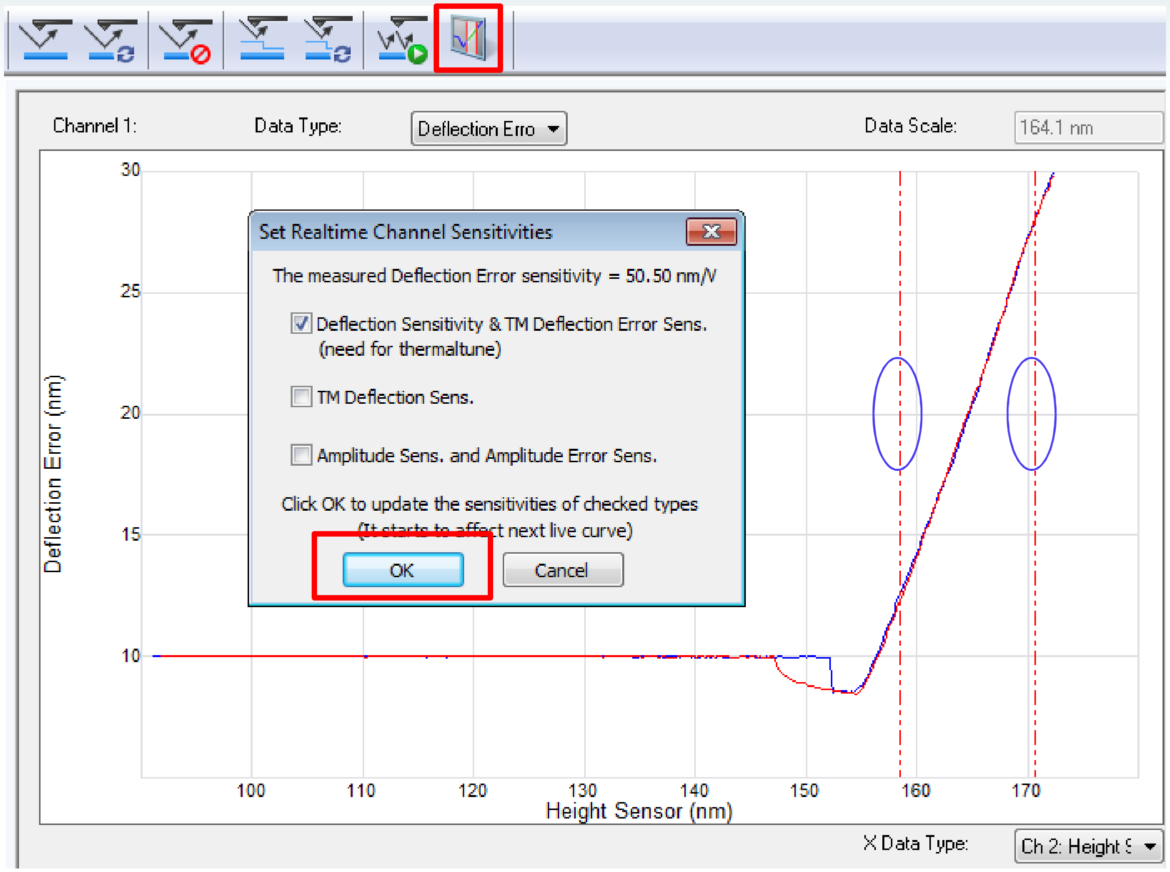
Def. Sens. Calibration Using Force Ramp

- Open PeakForce QNM In Air experiment.
- Load fused silica or sapphire sample.
- Engage
- **Scan 1um area to make sure center of the image (where is ramp will be performed) is flat and clean. Offset and/or zoom if needed.**
- Switch to Ramp Mode
- Configure ramp params
- Trigger Threshold
  - ScanAsyst Air or Tap150A - .5V
  - RTESPA or Tap525 - .2V
- Reduce the ramp size to get enough data points in contact part of the curve
- Analysis data in either Nanoscope or Nanoscope Analysis



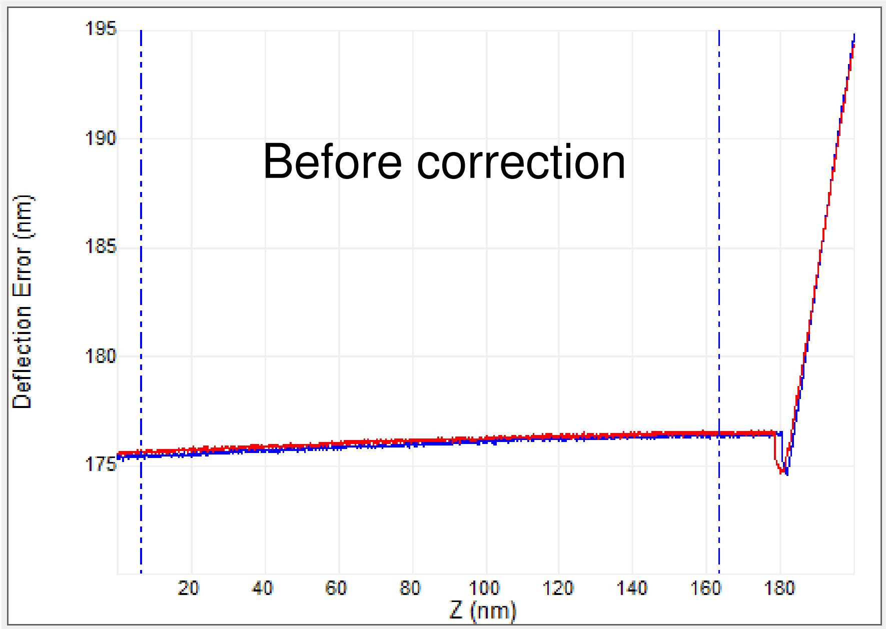
- **Question:** Can HOPG (17GPa) be used for Def. Sens. Calibration?
- **Question:** Can we calibration Def. Sens. in air, and then use it in fluid?

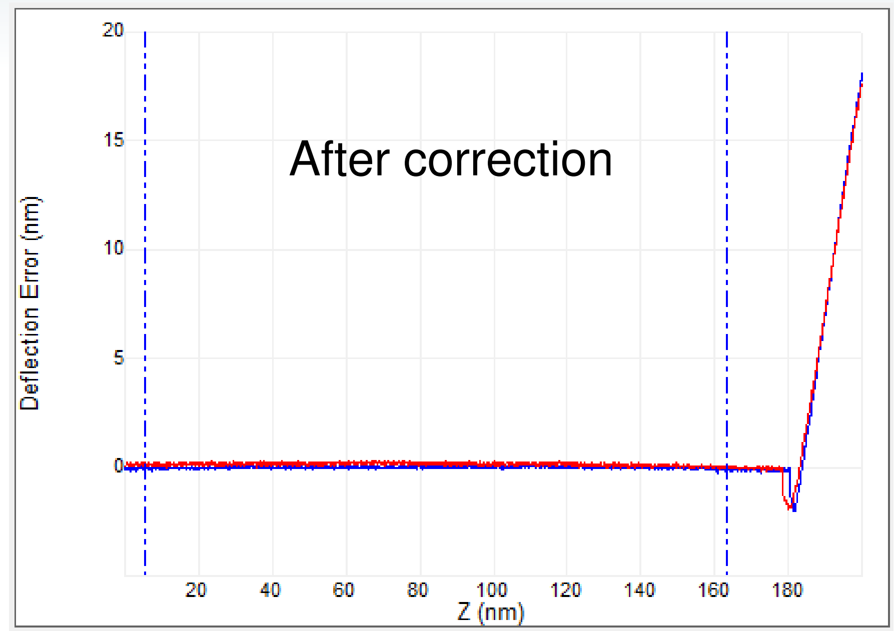
Def. Sens. Cal. in NanoScope

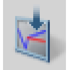


- Deflection Sensitivity can be calibrated in Nanoscope software
- After click “OK”, it will be updated in software automatically

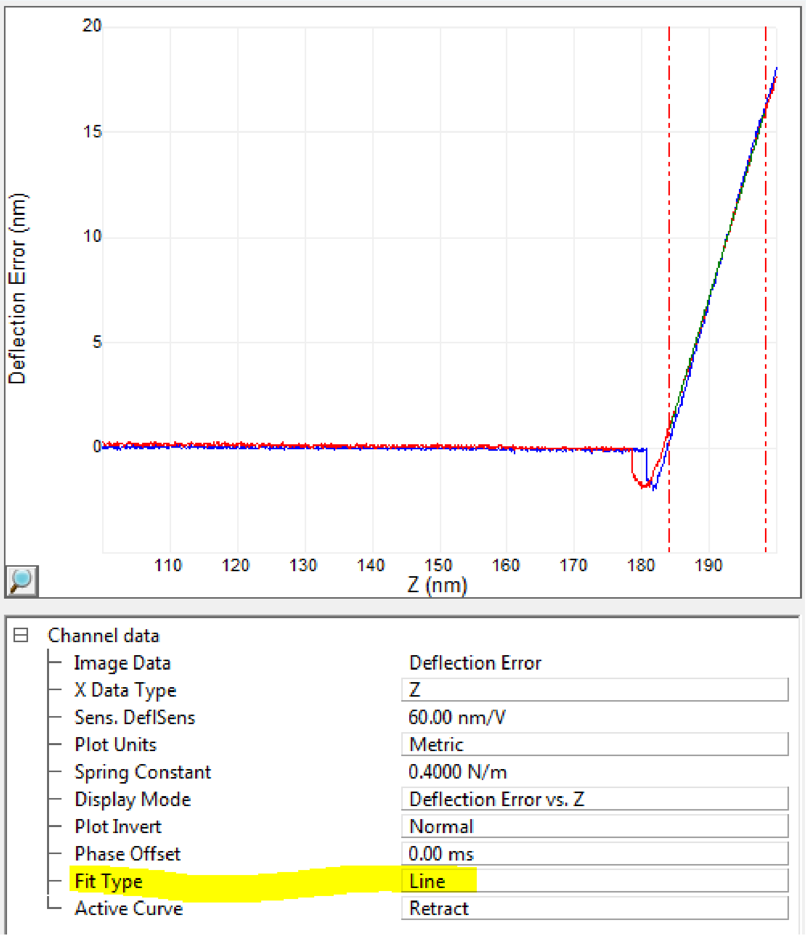
Def. Sens.: Baseline Correction

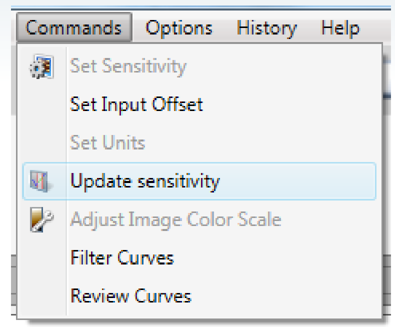


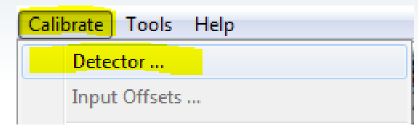


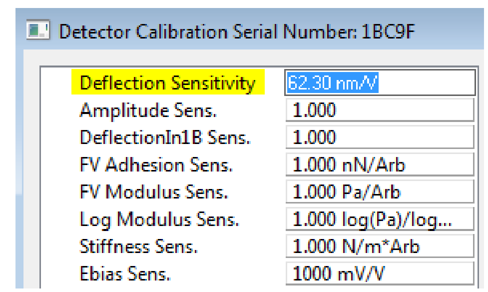
- Because of the optical interference or long range force (magnetic or electric static force), the baseline of the force curve may not be flat.
- Save the ramp curve, and open it in Nanoscope Analysis
- Click on  , define baseline fit region, then correct the baseline

Def. Sens.: Update Sensitivity with Line Fit



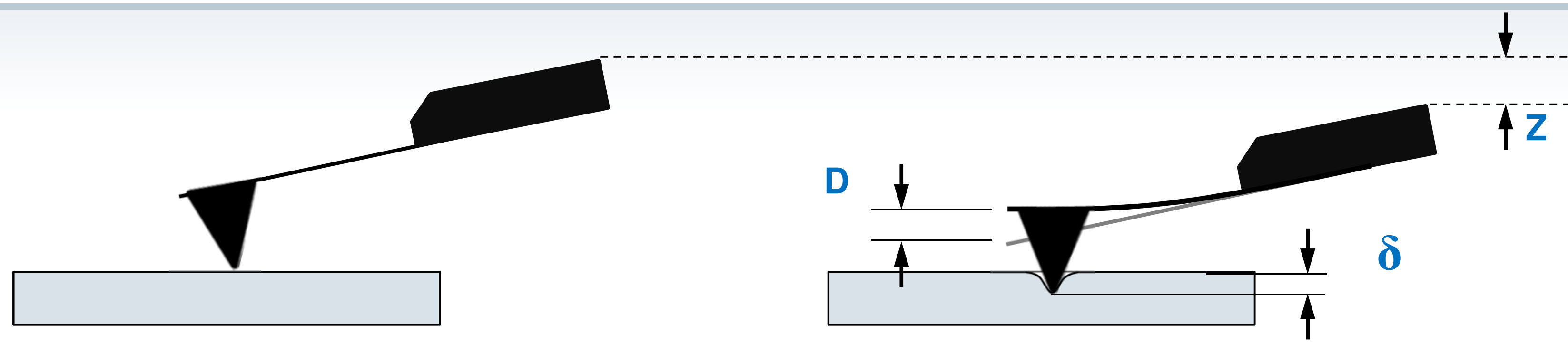






- Acquire at least 3 ramp curves, and calculate the average deflection sensitivity, then type manually in Calibrate window
- Confirm deflection sensitivity:
  - If it is a standard probe, compare with factory value
  - Deformation on hard sample should be close to zero
  - If not, click “autoconfig”

Deflection Sensitivity Error Transfers into Indentation Depth Error



- Z scanner moves: **Z**
- Cantilever Deflection: **D**
- Indentation depth: **δ=Z-D**
- Deflection Sens. calibration error (relative, %): **e**
- Measured Deflection: **(1+e)D**
- Calculated Indentation depth: **Z-(1+e)D**
- Indentation depth error (relative,%):**eD/(Z-D)=e(D/δ)**

- The Indentation Depth error (%) introduced by Deflection Sensitivity error could be amplified by a factor of **(D/δ)**
- **Question:** How to prevent Deflection Sensitivity calibration error from being amplified during indentation depth calculation?

PF-QNM Calibration

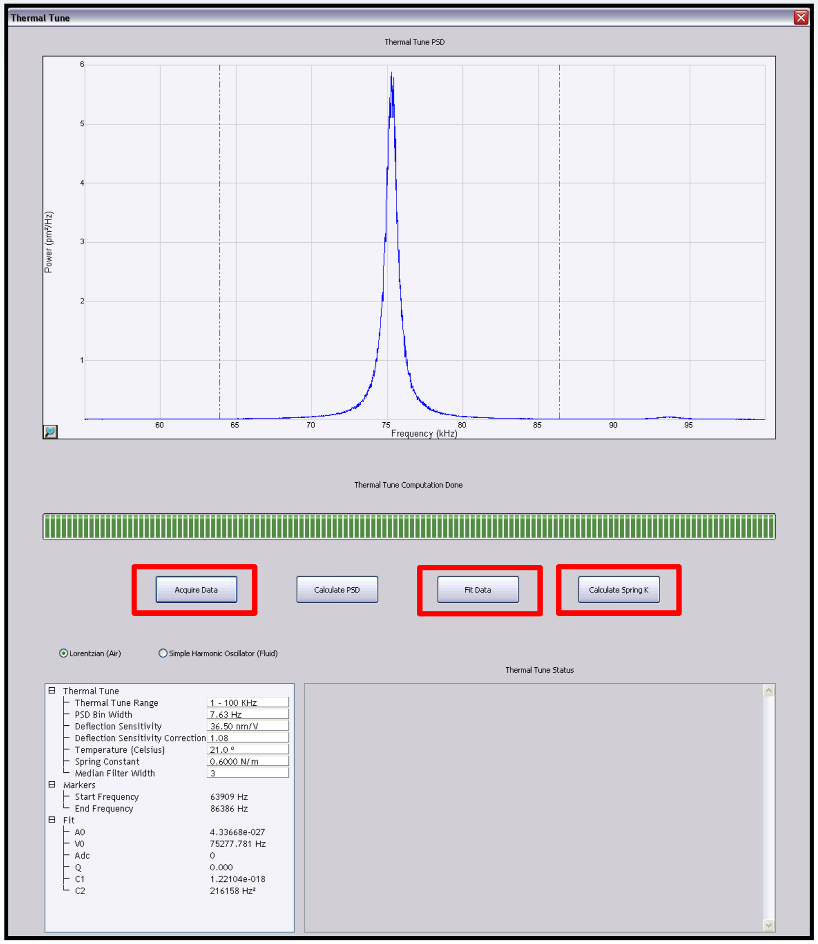
1. Deflection Sensitivity
2. **Spring Constant**
3. Tip Radius

Spring constant: Thermal Tune, K<1N/m

$$k = \frac{0.971 k_B T}{\chi^2 \langle Z_1{}^{*2} \rangle}$$

k, Spring constant
T, Temperature
k_B, Boltzmann constant, ($1.38 \cdot 10^{-23}$ J/K)
$Z_1{}^{*2}$, Mean square cantilever displacement
χ, Deflection sensitivity correction factor, 1.09

- **First make sure deflection sensitivity and thermal tune gains are calibrated corrected.**
- Withdraw the probe to prepare for thermal tune.
- Click the Thermal Tune Icon.
- Click the "Acquire Data" button.
- Drag in cursors for the data fit.
- Click the "Fit Data" button.
- Click the "Calculate Spring k" button.
- Click Yes to accept the k value.

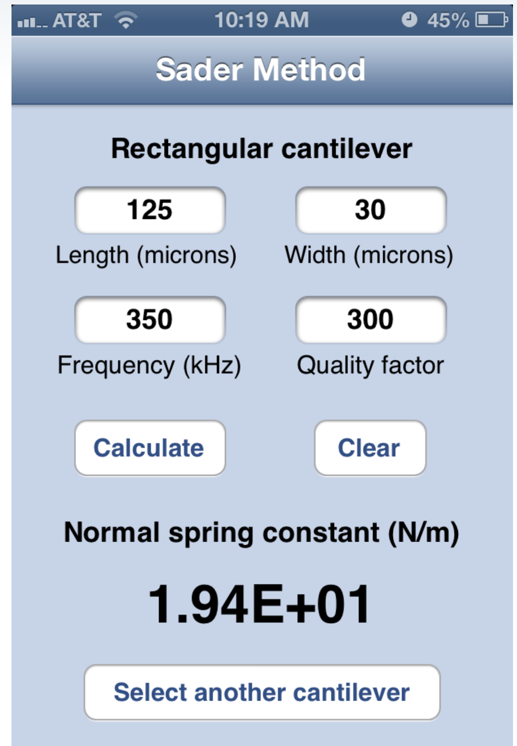


- **Question:** If we know cantilever's spring constant, can we use thermal tune to calibrate Deflection Sensitivity?

Spring Constant: Sader Method

$$k = 7.5246\rho_f w^2 L Q f_0^{\,2} \Gamma_i(Re)$$

$$Re = \frac{2\pi\rho_f f_0 w^2}{4\eta_f}$$



- Sader Method: rectangular shape (*free iPhone App "Sader Method"*)
- Length and Width can be optically measured with good accuracy
- Resonance frequency and Q factor can be obtained from thermal (recommend) or mechanical tune

Spring constant: other methods

- Dimensional Models
  - Dimensional model Simple Beam or Frequency scaling: Rectangular shape

$$k = \frac{Ewt^3}{4L^3}$$

Simple Bean Method

$$k \approx \frac{2\pi^3 w\left(f_0 L\sqrt{\rho}\right)^3}{\sqrt{E}}$$

Frequency Scaling Method

- Static Deflection Measurements
  - Reference cantilever method: $0.3K_{ref}<K<3K_{ref}$, all shape
  - Bruker supplies calibrated reference probe, model: CLFC-NOBO, which consists three cantilevers with nominal spring constant at: 0.16N/m, 1.3N/m, and 10.4N/m, respectively

Application Notes_094

Uncertainty of Spring Constant Calibration

Table 2: Overall uncertainty in spring constants

Method	Uncertainty	Main source of error
Simple beam, Eqn. (1)	~16%	Cantilever thickness
PBA, Eqn. (2)	~26%	Elastic modulus of SiN
Freq. scaling, Eqn. (4)	~9%	Si density
Reference cantilever, Eqn. (5) and (6)	~9%	Deflection sensitivity
Added mass, Eqn. (9c) and (10)	15-30%	Particle diameter
Sader, Eqn. (11)	~4%	Cantilever width
Thermal tune, Eqn. (15a)	~8%	Deflection sensitivity

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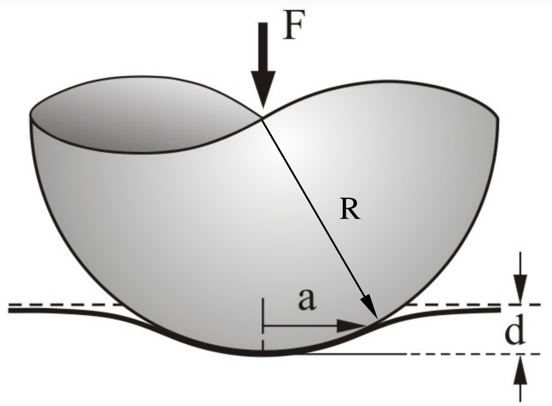
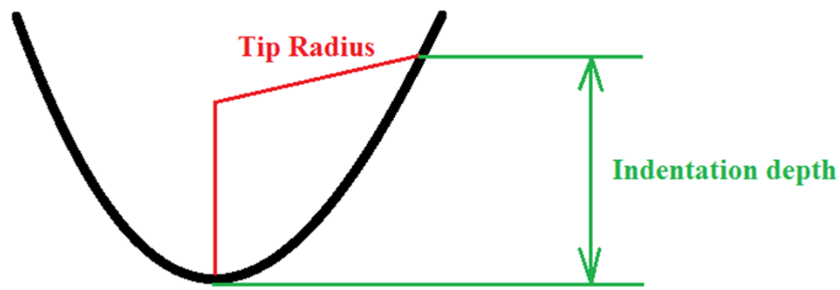
PF-QNM Calibration

1. Deflection Sensitivity
2. Spring Constant
3. **Tip Radius**

Tip Radius for DMT Model

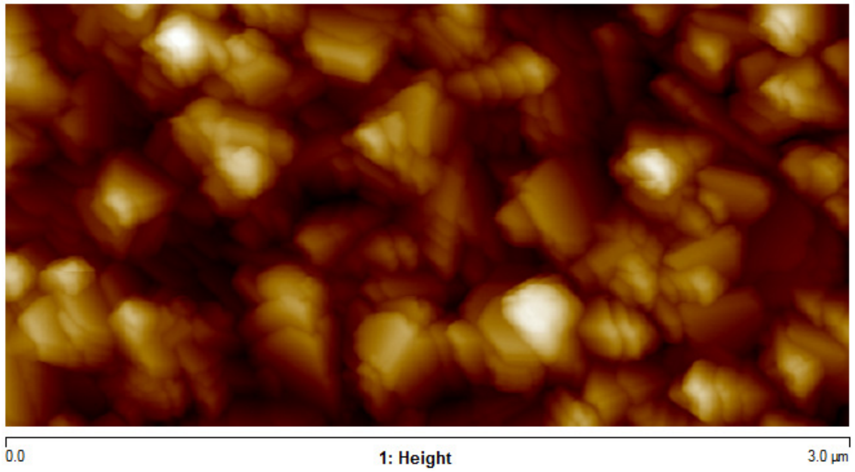




- For spherical probe, the tip radius is independent to the indentation depth
- For conical probe that tip end is not a part of a sphere, the effective tip radius depends on indentation depth
- Two information needed to calibrate tip radius:
    1. Tip shape
    2. Indentation depth

Determining Tip Radius with Tip Qual

Scan	
Scan Size	3.00 µm
Aspect Ratio	2.00
X Offset	0.000 nm
Y Offset	0.000 nm
Scan Angle	0.00 °
Scan Rate	0.501 Hz
Tip Velocity	3.00 µm/s
Samples/Line	512
Lines	256
Slow Scan Axis	Enabled
Feedback	
Peak Force Setpoint	1.890 nN
Feedback Gain	14.92
LP Deflection BW	40.00 kHz
ScanAsyst Noise Threshold	1.00 nm
ScanAsyst Auto Control	Individual
ScanAsyst Auto Gain	On
ScanAsyst Auto Setpoint	On
ScanAsyst Auto Scan Rate	Off
ScanAsyst Auto Z Limit	Off
PeakForce QNM Control	
Peak Force Amplitude	150 nm
Lift Height	23.2 nm

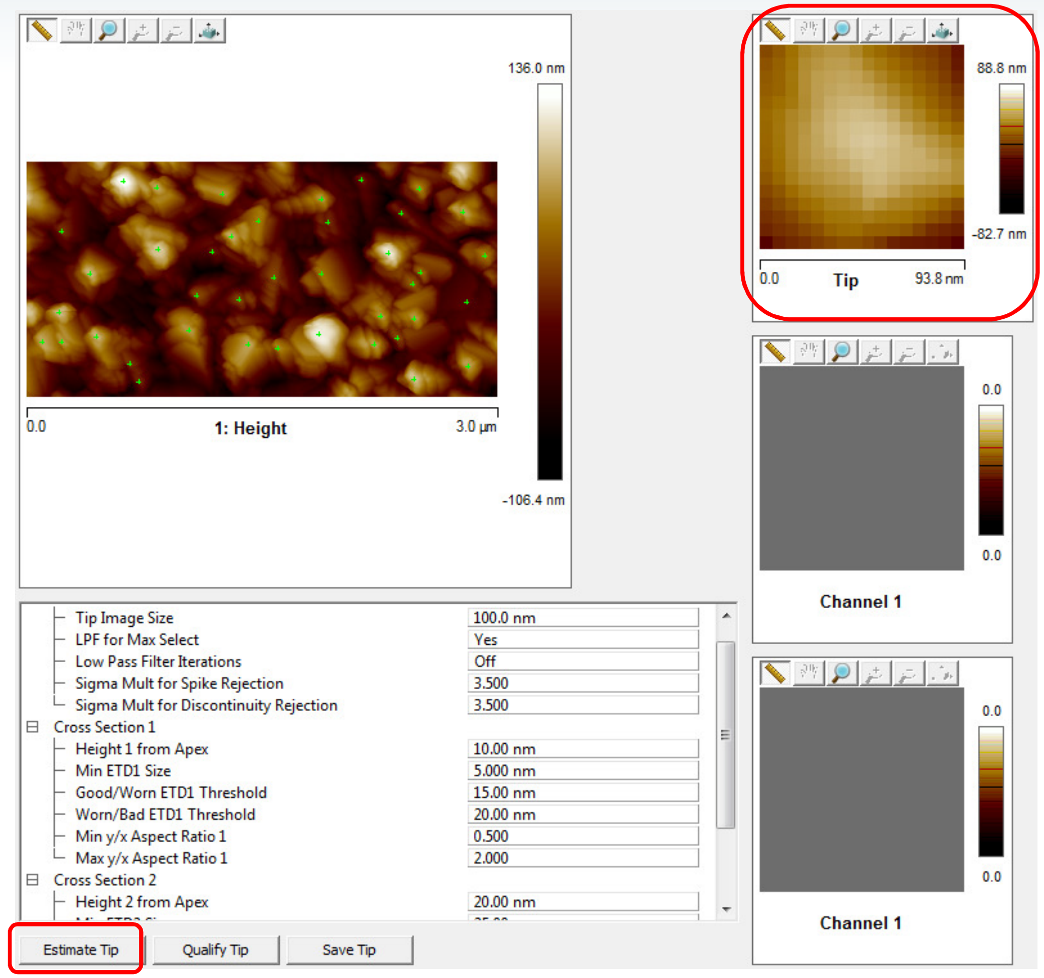


- Acquire an Tip Check sample image with parameters as shown
- Open the captured image with NanoScope Analysis
- Set Plane Fit Mode to 1st order XY plan fit
- **Question:** How to use DNISP diamond probe (K=225N/m) to image tip check sample?

Determining Tip Radius with Tip Qual

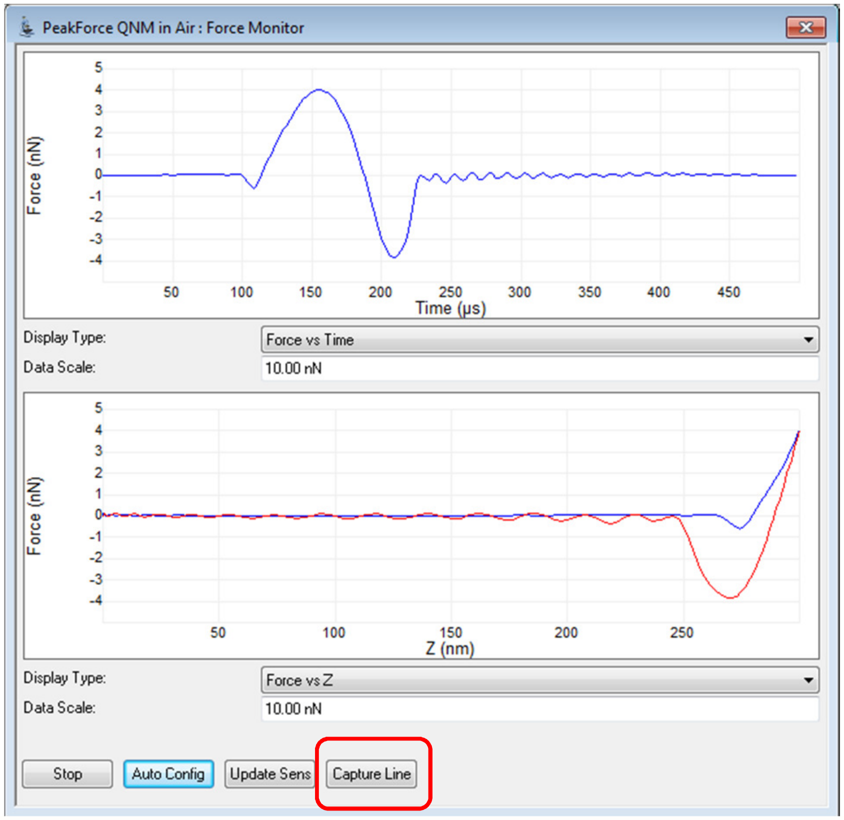
Tip Qual Icon

- Click the Tip Qual Icon
- Set Tip Image Size to 100 nm
- LPF for max select On
- Click on Estimate Tip
  - An image of the Tip now appears in the image window



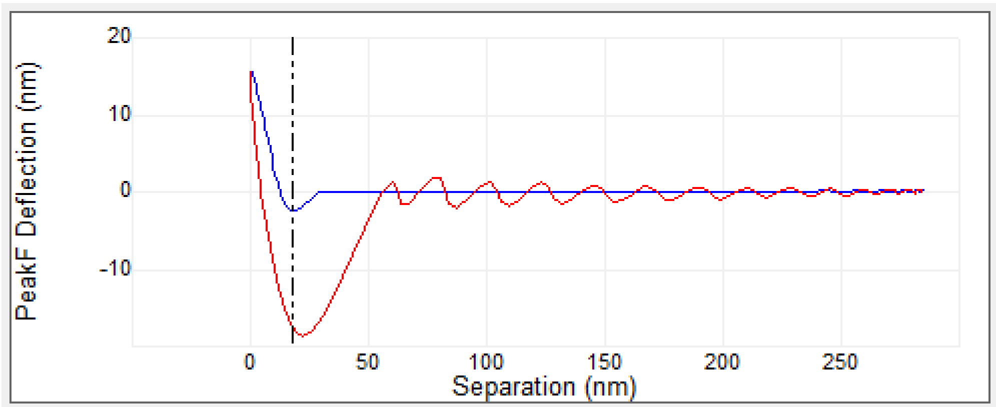
PFQNM – How To – HSDC & PFQNM Offline (how to determine true indentation depth)

- Capture an image on the unknown sample.
- While the image capture is in progress press the Capture Line button on the Force Monitor to capture lines of raw data.
- Click Upload Data button to save the raw HSDC data to a file.



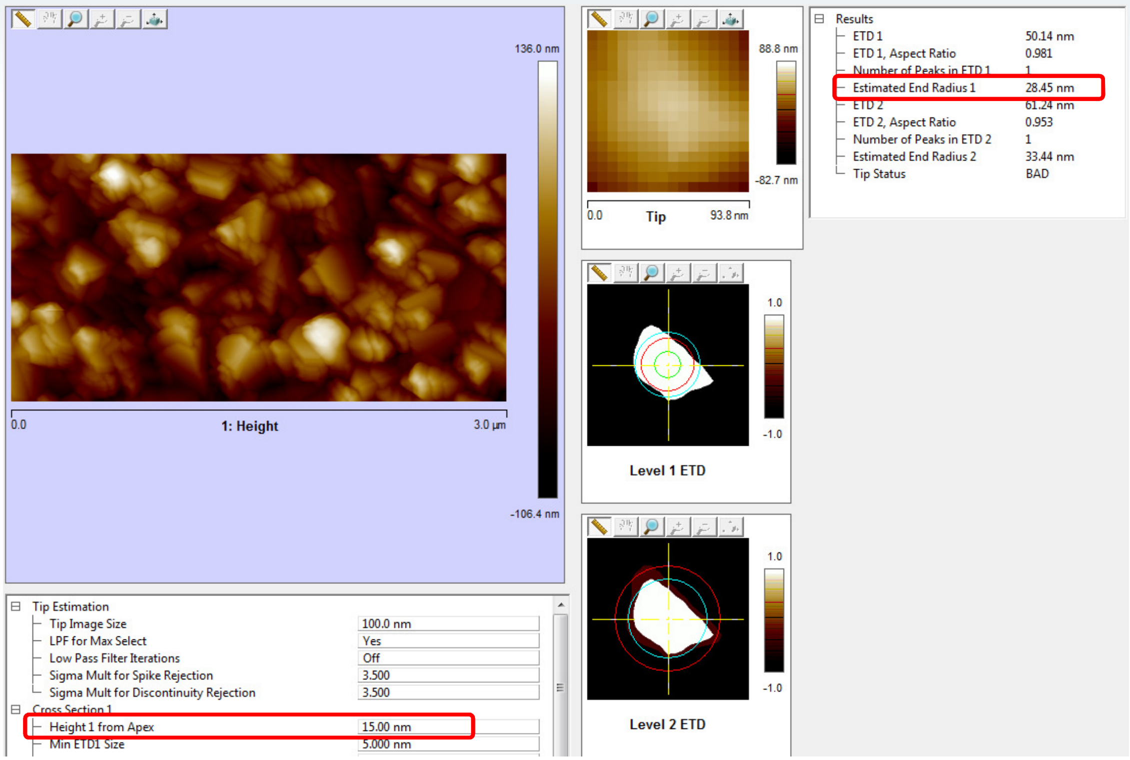
Indentation depth : Deflection vs. Separation plot of HSDC

- The distance on a Deflection vs. **Separation** plot's load (extend) curve starting at the "snap-to-contact" point and ending at z fully extended.
  - Use HSDC and QNM HSDC-ForceCurve analysis to export force curve(s) and then reload to display separation plot
  - The force monitor Force vs. Z is a good approximation only if the cantilever deflection is small compared to the deformation



Indentation from deflection vs. separation plot load curve

Determining Tip Radius with Tip Qual

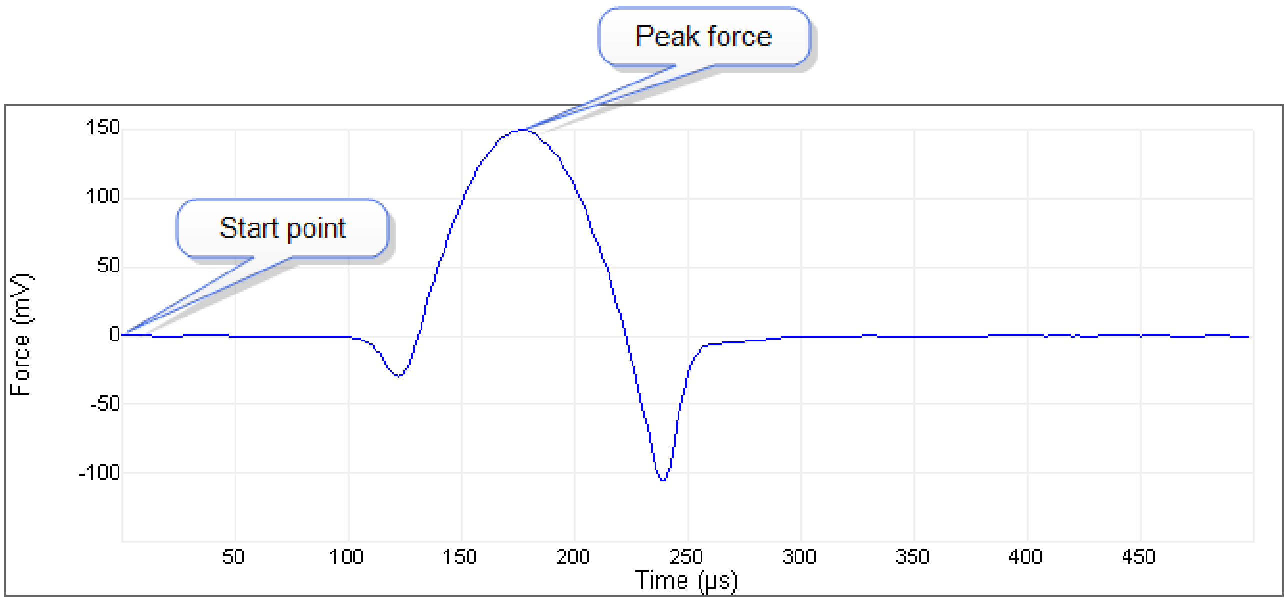


- Enter the average indentation depth into Height 1 or Height 2 From Apex
- Click Qualify Tip
- Read the corresponding Estimated End Radius (not ETD or ETD/2)
- Enter Tip Radius into "Cantilever Parameters / Tip Radius" in real time software
- **Question:** If we use a 5um diameter sphere probe, how to calibrate the tip radius?

Other Parameters

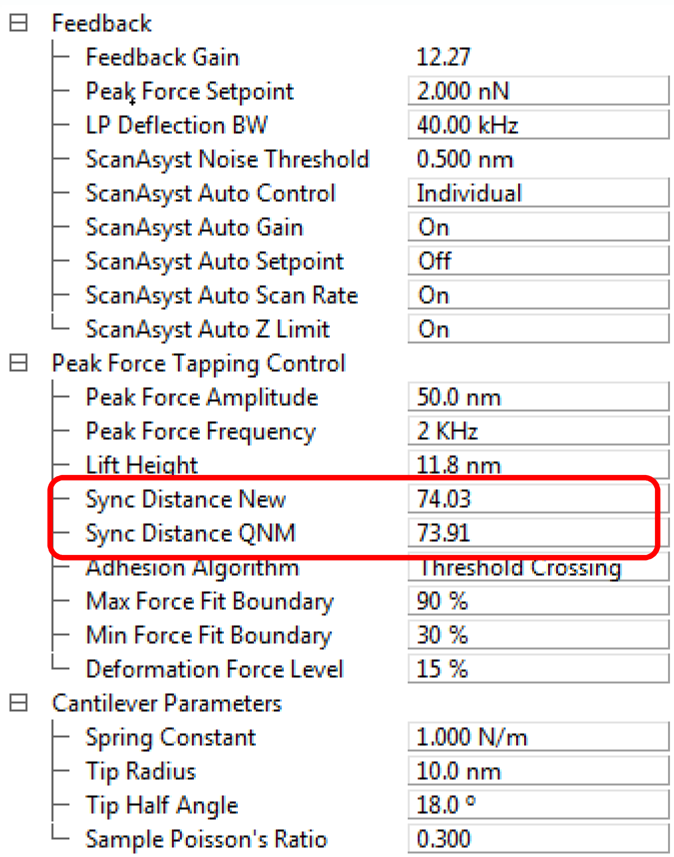
1. Sync Distance
2. Drive3 Amp. Sens.

1. QNM Sync Distance Calibration (What is Sync Distance)



- Sync Distance is a time constant during each tapping cycle when cantilever reaches the lowest position.
- On hard sample deflection reaches max when piezo extends to lowest position.
- In QNM analysis, Sync Distance or PeakForce position is used as the turning point between approach force curve and retract force curve.

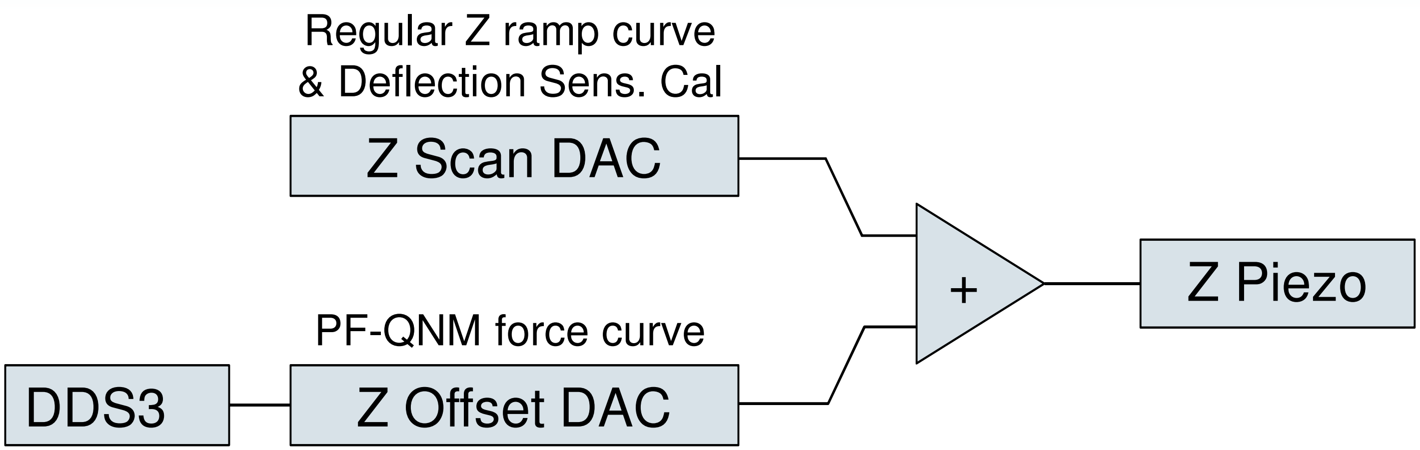
1. QNM Sync Distance Calibration (V9 or later Nanoscope software)



If it is difficult to determine the Sync Distance for the unknown sample, do the following:

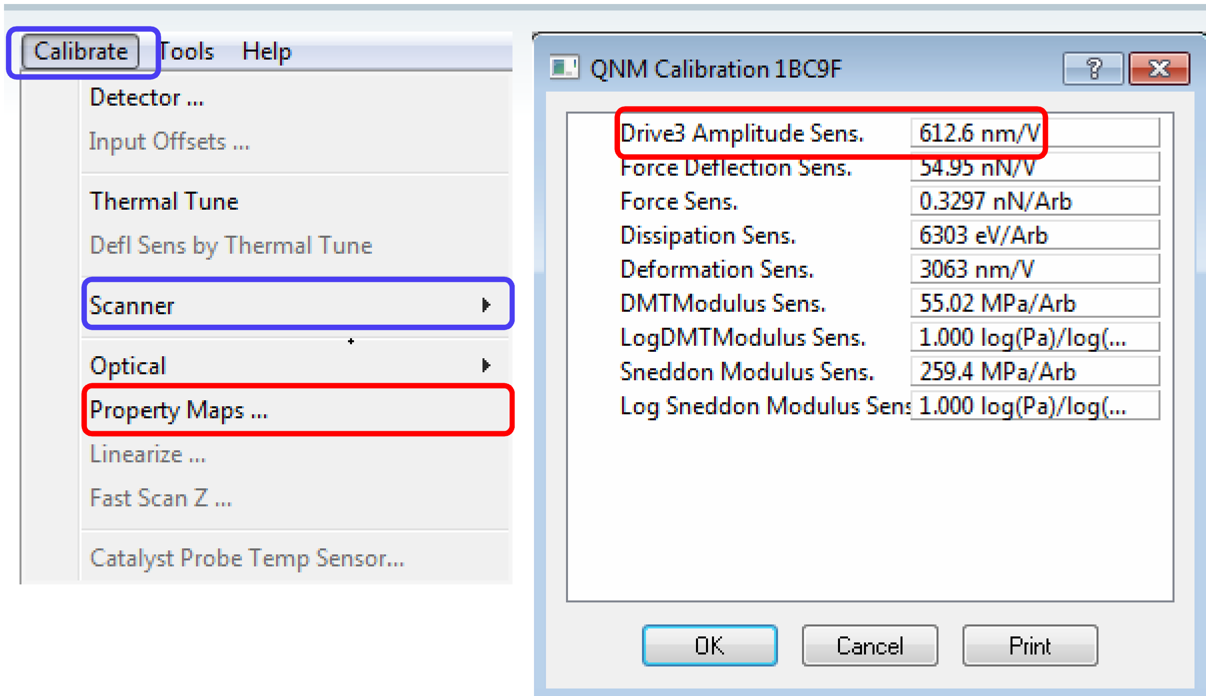
- There are two Synch distance (V9)
  - Sync Distance New (for imaging only)
  - Sync Distance QNM
- Use the same probe to calibrate the Sync Distance QNM on a hard elastic sample use auto config, and write it down. (may do this right after deflection sensitivity calibration)
- Manually type this Sync Distance QNM in Nanoscope software for the unknown sample

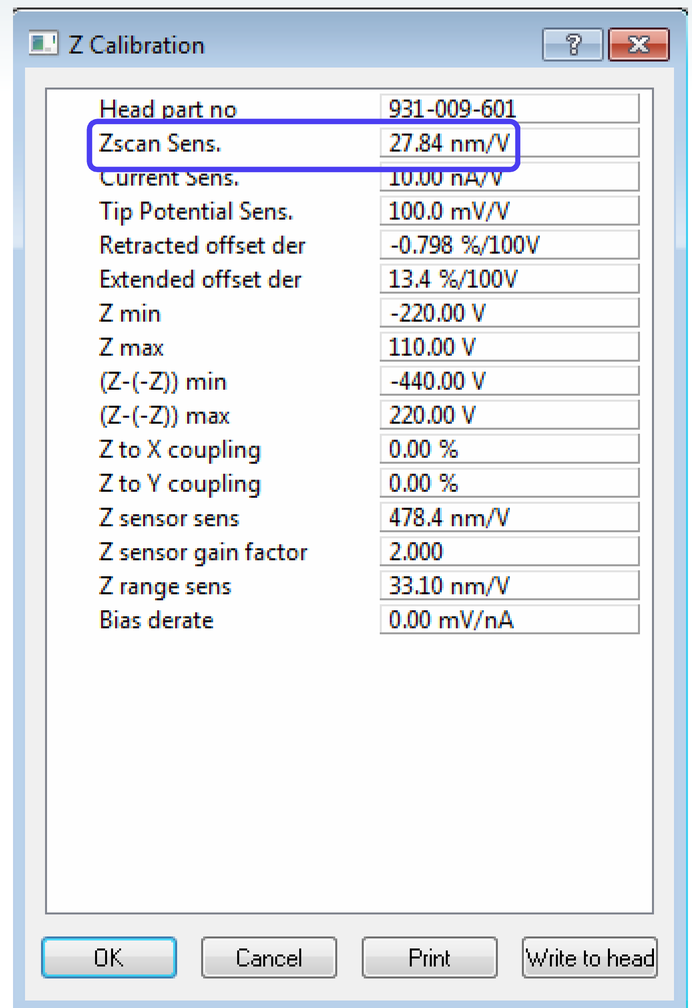
2. Drive3 Amplitude Sensitivity Calibration (how the modulation is driven in QNM)



- Two reasons why need Drive3 Amp. Sens. Calibration:
  1. Deflection sensitivity calibration (regular Z ramp) and PF-QNM force curve uses different DAC to generate the drive signal
  2. In some case (e.g. Icon Scanner at 2KHz), the Drive3 amplitude sensitivity maybe frequency dependent, and can be calibrated or manually adjusted

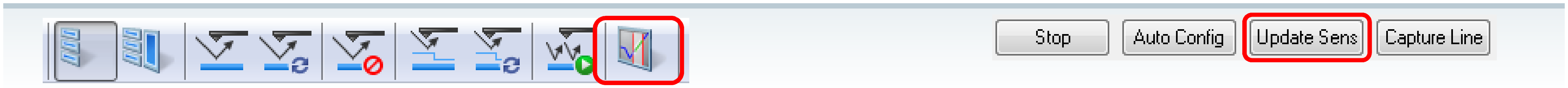
2. Drive3 Amplitude Sensitivity Calibration (What is Drive3 Amp. Sens.)

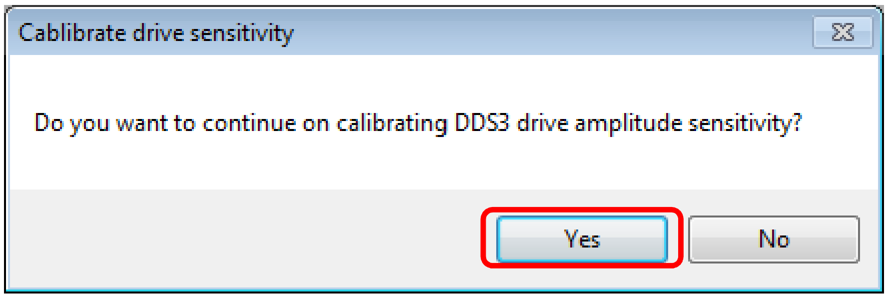


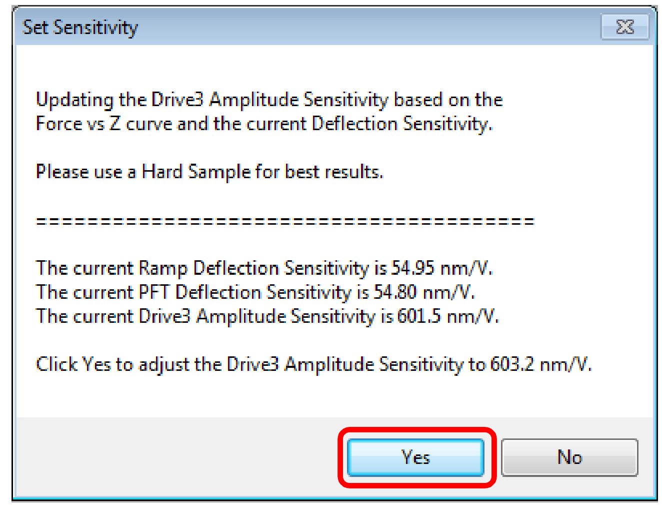


- By default, the Drive3 Amplitude Sensitivity is calculated from Z scan sensitivity
- But Drive3 Amplitude Sens. Can be independently calibrated or manually typed in
- Reloading the workspace or software will set Drive3 Amplitude Sens. to default value

2. Drive3 Amplitude Sensitivity Calibration (V9 and Later Nanoscope software)

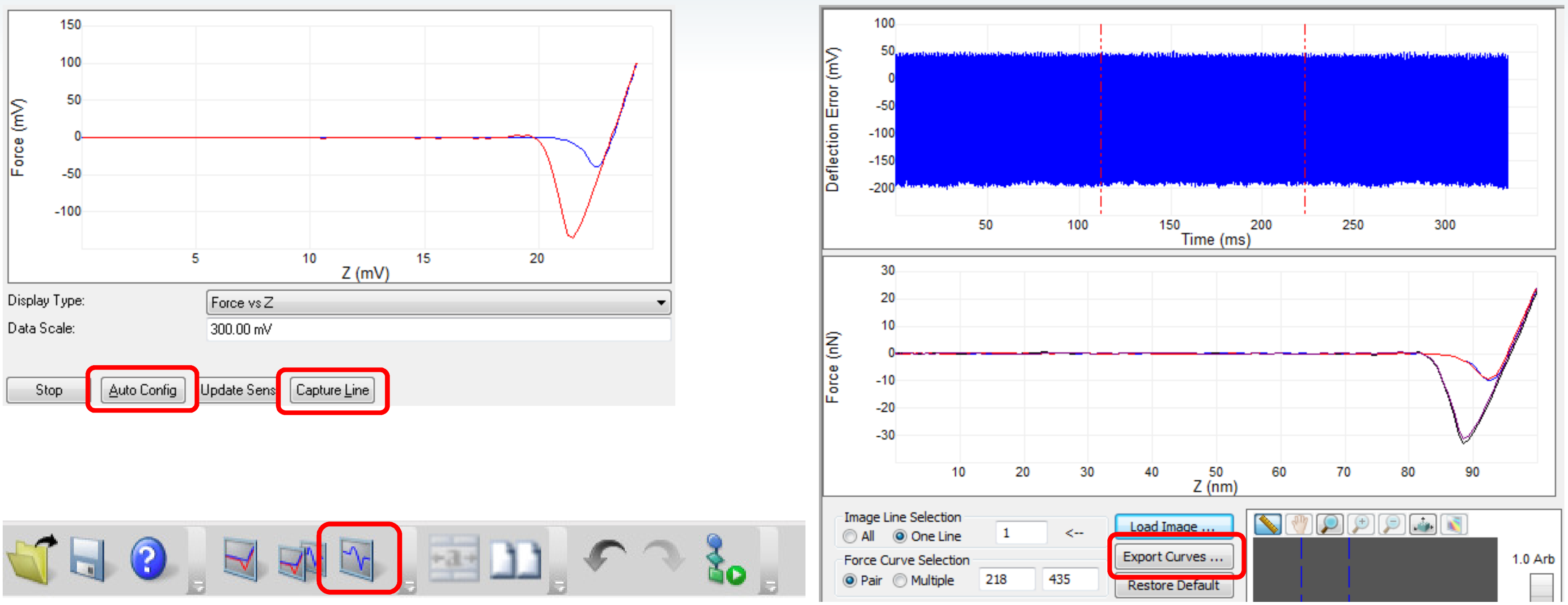






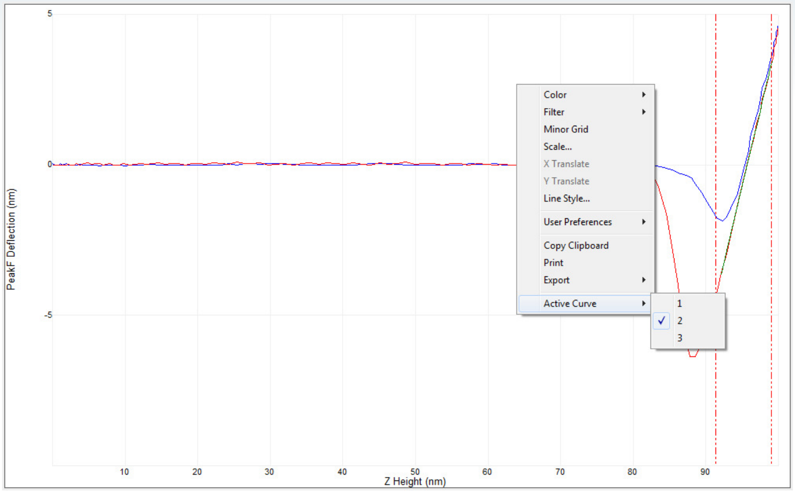
- Two ways to calibrate the Drive3 Amplitude Sensitivity:
  1. In ramp mode: get ramp curve and update deflection sensitivity in Nanoscope SW, then calibrate DDS3 drive amplitude sensitivity when message pops up
  2. After measuring the deflection sensitivity in ramp mode, return to scan mode and with PeakForce setpoint>0.1V, click “Update Sensitivity” in PF-QNM force window

2. Drive3 Amplitude Sensitivity Calibration (V8.15 and Later Nanoscope software)



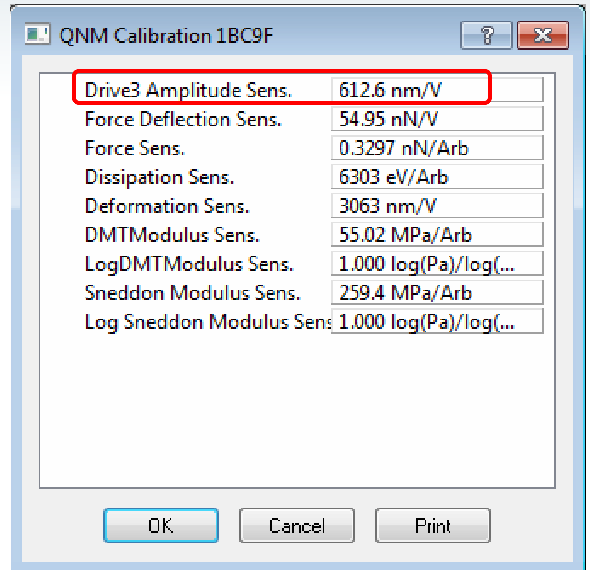
- Make sure deflection sensitivity is already calibrated
- PeakForce Tapping on hard surface (same sample used for Def. Sen. calibration)
- Set PeakForce Setpoint to 0.1-0.2V
- Click "Auto Config" to make sure Sync Distance is correct
- Click "Capture Line" to Capture force curve
- Open the HSDC data in Nanoscope Analysis, and export a force curve

2. Drive3 Amplitude Sensitivity Calibration (V8.15 and Later Nanoscope software)



Plot Units	Metric
Spring Constant	5.000 N/m
Display Mode	PeakF Deflection vs. Ch 2:
Plot Invert	Normal
Phase Offset	0.00 ms
Fit Type	Line
Active Curve	Retract

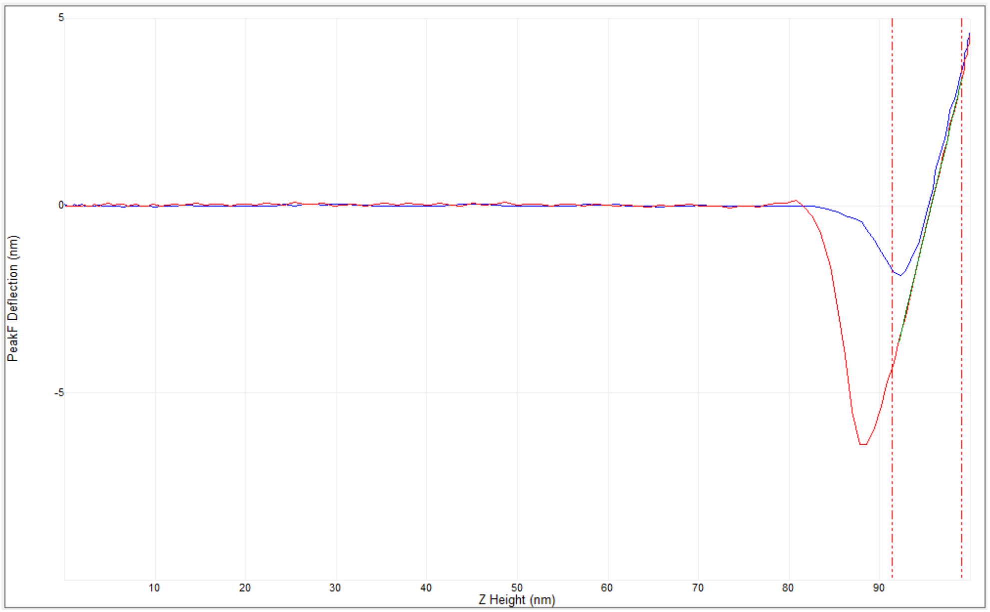
dY/dX	1.0577 nm/
dX/dY	0.9454 /nm
Least Squares Equation	y = (1.0763 nm/)x - 103.0nm
Root Mean Square	0.07715 nm
Average Deviation	0.06510 nm



New calibrated Drive3 Amp. Sen.
=612.6nm/V*1.0763
=659.3nm/V

- Open the saved force curve in Nanoscope Analysis; line fit the contact region of the retract curve, and get the slope of the fitted curve (e.g. 1.0763)
- Click “Calibrate”>>”Property Maps” to find the current Drive3 Amp. Sens.
- New “Drive3 Amp. Sens.”=“fitting slope”*current “Drive3 Amp. Sens.”
- Type new “Drive3 Amp. Sens.” (e.g. 659.3nm/V) to “Property Maps”, and click “OK”

2. Drive3 Amplitude Sensitivity Calibration (V8.15 and Later Nanoscope software)



Marker Pair 1	
X0	91.54
X1	99.10
Y0	-4.178 nm
Y1	3.431 nm
X1 - X0	7.561
Y1 - Y0	7.609 nm
dY/dX	1.0064 nm/
dX/dY	0.9937 /nm
Least Squares Equation	y = (1.0135 nm/)x - 96.9945nm
Root Mean Square	0.06599 nm
Average Deviation	0.05228 nm

- Do a new Capture Line, then open the HSDC data in Nanoscope Analysis and export a representative force curve
- Line fit the contact part of the exported force curve, now the slope should be close to 1.0
- Now "Drive3 Amp. Sens." is calibrated

Relative Method vs. Absolute Method

$$DMT\ Model: \quad \boldsymbol{F = Def.* K = \frac{4}{3}\frac{E}{(1-\nu^2)}R^{1/2}\delta^{3/2}}$$

- E~ k/√R (E=modulus, k=spring constant, R=tip radius)
- Main differences between "relative" and "absolute" methods.
  - Absolute method requires direct measurement of k and R
    - R must be evaluated at indentation depth
    - If indentation depth changes R can be re-evaluated, but modulus is only weakly dependent on R
  - Relative method uses reference sample to force the *ratio* of k/√R to be correct at a given indentation depth.
    - To get modulus of unknown sample adjust setpoint until the indentation depth is close to that used to establish k/√R on the reference sample.
- Reference sample should have similar modulus as unknown sample
- **Question:** how to make sure the indentation depth is close to that on reference sample?

PF-QNM Procedure

Absolute Method

1. Calibrate Deflection Sensitivity
2. Calibrate Drive3 Amp Sensitivity and QNM Sync Distance
3. Calibrate Spring Constant
4. Image on the tip check sample, if tip radius needs to be calibrated
5. Switch to the unknown sample
6. Manually adjust the PeakForce setpoint to get 5-10nm deformation
7. Capture HSDC data on unknown sample
8. Measure Indentation Depth
9. Open tip check image in Nanoscope Analysis, and use Indentation depth to get the tip radius
10. Type in the tip radius in Nanoscope software and get the sample modulus

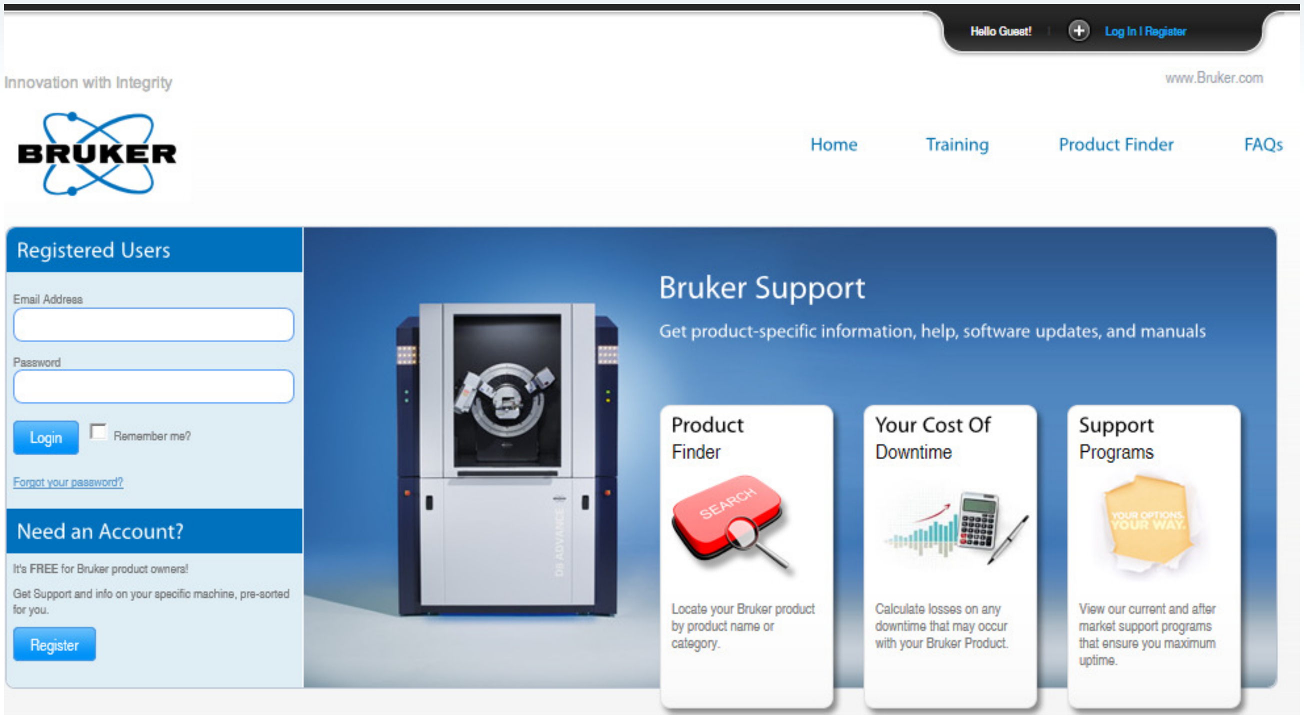
Relative Method

1. Calibrate Deflection Sensitivity
2. Calibrate Drive3 Amp Sensitivity and QNM Sync Distance
3. Load a reference sample with known modulus
4. Manually adjust the PeakForce setpoint to get 5-10nm deformation
5. Type in an estimated spring constant (i.e. nominal value)
6. Adjust the tip radius so that the measured modulus is close to expect value
7. Record the average deformation
8. Switch to unknown sample
9. Only adjust the PeakForce setpoint so that the average deformation is same as on reference sample
10. The measured modulus is the unknown sample modulus

Summary

- Three basic Calibrations: Deflection Sensitivity, Sprint Constant, Tip Radius
- Two advanced Calibrations: Drive3 Amp. Sensitivity, Sync Distance QNM
- Cantilever spring constant needs to match the sample modulus, relative stiff cantilever is better
- Deflection Sensitivity is the most important parameter in QNM calibration
- Make sure sync distance is correct
- Check Drive3Amp sensitivity, especially when use 2 KHz frequency
- Manual control PeakForce setpoint for QNM
- Need to have enough Deformation (at least 2-5nm)
- Relative method is preferred for hard material (>1GPa)

Bruker AFM Technical Support



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